Jinyu Xie

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Education

2021 – Present **B.S., UESTC** in Mathematics-Physics Fundamental Science
Yingcai Honors College (Elite program, top 102 students selected from 5000 freshmen).
GPA: 3.89/4.00; Avg. Score: 89.13/100.00.

Research Publications

1. **J.-Y. Xie**, H. Zhang, X. Zhao, and Y. Luo, "IRTF: A new tensor factorization for irregular multidimensional data recovery," *Knowledge-Based Systems*, under review.
2. **J.-Y. Xie***, J.-Y. Li *, Y.-S. Luo, X.-L. Zhao, and J.-L. Wang, "H2TF for hyperspectral image denoising: Where hierarchical nonlinear transform meets hierarchical matrix factorization," *IEEE Geoscience and Remote Sensing Letters*, vol. 20, 2023. DOI: 10.1109/LGRS.2023.3294933.
3. Y.-Y. Liu, X.-L. Zhao, **J.-Y. Xie**, X. Zhi, and G. Vivone, "Bi-level tensor decomposition for hyperspectral image restoration," in *IEEE International Geoscience and Remote Sensing Symposium*, 2024.
4. H. Zhang, T.-Z. Huang, X.-L. Zhao, S. Zhang, **J.-Y. Xie**, T.-X. Jiang, and M. K. Ng, "Learnable transform-assisted tensor decomposition for spatio-irregular multidimensional data recovery," *ACM Transactions on Knowledge Discovery from Data*, forthcoming.

Research Experience

2023/07 – present **Topic: Tensor Decomposition for Irregular Data**. Supervised by Prof. Xile Zhao.
I studied irregular data algorithms and developed a novel tensor decomposition method for irregular data (e.g., spatial transcriptomics data) reconstruction. A paper is submitted based on this work.

2022/07 – present **Topic: Advanced Tensor Decomposition**. Supervised by Prof. Xile Zhao.
I have studied tensor decomposition algorithms for various data modeling and, in collaboration with others, have developed a novel tensor decomposition method that integrates deep learning techniques. Based on this work, I have published a journal article.

2022/01 – 2022/07 **Topic: Deep Learning Algorithms Research**. Supervised by Assoc. Prof. Liangjian Deng.
I studied convolutional neural network (CNN) algorithms for super-resolution and deraining and collaborated on an orange quality classifier using CNNs. I then learned spiking neural network (SNN) theories and worked on implementing SNNs with memristors and mathematical modeling of the combination.

Patents

2024/04 H. Zhang, T.-Z. Huang, X.-L. Zhao, S. Zhang, J.-Y. Xie, T.-X. Jiang, and M. K. Ng. "Irregular high-dimensional data restoration method and system based on transform domain tensor decomposition". CN117911802A, filed January 17, 2024, and issued April 19, 2024.

Awards

2023/03 Second Prize, 14th Mathematics Competition of Chinese College Students (Mathematics Class A).
2023/12 **Excellent**, University Student Innovation and Entrepreneurship Competition (continuous funding).
2023/05 **First Prize**, 23rd Mathematical Modeling Contest, UESTC.
2022/12 **Outstanding**, 2021 Freshman Extracurricular Innovation Project, UESTC.
2023/06 **Outstanding**, University Student Innovation and Entrepreneurship Training Program, UESTC.
2024/6 **Exemplary Student Scholarship**, Exemplary Student Scholarship, UESTC (three times).

Skills

Coding Python, Matlab, LaTeX, Markdown.